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Title: Chemical Spill Cleanup Procedure

Chemical Spill Clean Up Procedure

This document should be read in conjunction with Chemical Safety procedure.

1. Warn other staff in the near vicinity of a spillage
2. Be aware of the dangers of the chemicals you are using. Refer to COSHH / SDS in the Hazard Manual for each Department. These will give a list of the safety information and first aid measures to be taken for all chemicals used in the department.
3. Collect spill kit. Each kit is contained in a white plastic bucket which has the following items:

   Plastic apron b. Face shield c. Nitrite gloves d. Spillex™ Products e. Plastic bags f. brush and shovel g. pH paper 0-14 h. Presept™ granules (biological spills)

Procedure:

a. When a spill has occurred, first decide whether you know what the chemical involved is or not. Call the safety officer or supervisor.
b. Consult material data safety sheets (COSHH) available in the hazard register. These will alert you to the dangers and tell you what precautions you need to take.
c. Be aware that the neutralization process maybe be exothermic and /or produce hazardous fumes. Call safety officer.
d. Safety equipment must be worn during the cleanup process
e. All spills must be neutralized with the appropriate Spillex™ product
   1. Spillex A Used for acids
   2. Spillex C Used for alkalis / caustic material
   3. Spillex XS Used for solvents
   4. Spillex FP Used for Formalin

Starting from the outside of the spill, pour the Spillex™ slowly around the spill in decreasing circles until all the liquid is absorbed and the pH is close to 7.0. Check with pH paper provided.

Sweep up absorbed liquid and place the slurry into the plastic bags provided. Plastic this into another yellow biohazard bag, together with gloves and apron, for disposal.

Concentrated Hydrochloric acid

Concentrated Nitric acid

Concentrated Sulphuric acid

Concentrated Diethyl Ether

Concentrated Formaldehyde

Concentrated Phosphoric acid

Ammonia

Chloroform

THE OSHA LABORATORY STANDARD

The basis for this standard (29 CFR 1910.1450) is a determination by the Occupational Safety and Health Administration (OSHA), after careful review of the complete rule-making record, that laboratories typically differ from industrial operations in their use and handling of hazardous chemicals and that a different approach than that found in OSHA's substance specific health standards is warranted to protect workers. The final standard applies to all laboratories that use hazardous chemicals in accordance with the definitions of laboratory use and laboratory scale provided in the standard. Generally, where this standard applies it supercedes the provisions of all other standards in 29 CFR, part 1910, subpart Z, except in specific instances identified by this standard. For laboratories covered by this standard, the obligation to maintain employee exposures at or below the permissible exposure limits (PELs) specified in 29 CFR, part 1910, subpart Z is retained. However, the manner in which this obligation is achieved will be determined by each employer through the formulation and implementation of a Chemical Hygiene Plan (CHP). The CHP must include the necessary work practices, procedures and policies to ensure that employees are protected from all potentially hazardous chemicals used or stored in their work area. Hazardous chemicals as defined by the final standard include not only chemicals regulated in 29 CFR part 1910, subpart Z, but also any chemical meeting the definition of hazardous chemical with respect to health hazards as defined in OSHA's Hazard Communication Standard, 29 CFR 1910.1200(c).

Among other requirements, the final standard provides for employee training and information, medical consultation and examination, hazard identification, respirator use and record keeping. To the extent possible, the standard allows a large measure of flexibility in compliance methods.

EMPLOYEE RIGHTS AND RESPONSIBILITIES

Employees have the right to be informed about the known physical and health hazards of the chemical substances in their work areas and to be properly trained to work safely with these substances.

Employees have the responsibility to attend training seminars on the Laboratory Standard and Chemical Hygiene Plan and to stay informed about the chemicals used in their work areas. They have the responsibility to use safe work practices and protective equipment required for safe performance of their job. Finally they have the responsibility to inform their supervisors of accidents and conditions or work practices they believe to be a hazard to their health or to the health of others.

HAZARDOUS CHEMICALS

The Laboratory Standard defines a hazardous chemical as any element, chemical compound, or mixture of elements and/or compounds which is a physical or health hazard.

A chemical is a **physical hazard** if there is scientifically valid evidence that it is a flammable, a combustible liquid, a compressed gas, an explosive, an organic peroxide, an oxidizer, pyrophoric, unstable material (reactive), or water-reactive.

A chemical is a **health hazard** if there is statistically significant evidence based on at least one study conducted in accordance with established scientific principles that acute or chronic health effects may occur in exposed employees. Included are:

- carcinogens
- irritants

- sensitizers
- neurotoxins (nerve)
- hepatotoxins (liver)
- agents that act on the hematopoietic system (blood)
- radioactive material
- biohazards
- nephrotoxins (kidney)
- agents that damage the lungs, skin, eyes, or mucous membranes

See Appendix I, Glossary, for definitions of these terms.

In most cases, the label will indicate if the chemical is hazardous. Look for key words like **caution, hazardous, toxic, dangerous, corrosive, irritant, carcinogen**, etc. Old containers of hazardous chemicals (before 1985) may not contain hazard warnings.

If you are not sure a chemical you are using is hazardous, review the **Safety Data Sheet (SDS)** or contact your supervisor, instructor, or the QA office.

Designated areas must be established and posted for work with certain chemicals and mixtures, which include **select carcinogens**, **reproductive toxins**, and/or substances which have a **high degree of acute toxicity**. A designated area may be the entire laboratory, an area of a laboratory or a device such as a laboratory hood. Designated area stickers are available from QA.

SAFETY DATA SHEETS (SDSs)

A Safety Data Sheet (SDS) is a document containing chemical hazard and safe handling information prepared in accordance with the OSHA Hazard Communication Standard. A sample SDS is included at the end of Part I.

Chemical manufacturers and distributors must provide a SDS the first time a hazardous chemical/product is shipped to a facility. Only SDSs received must be retained and made available to laboratory workers. However, you can request an SDS for any laboratory chemical from the manufacturer or distributor.

CHEMICAL INVENTORIES

The OSHA Laboratory Standard does not require chemical inventories; however, it is prudent to adopt this practice. An annual inventory can reduce the number of unknowns and the tendency to stockpile chemicals.

ROYAL HOSPITAL PATHOLOGY - CHEMICAL HYGIENE PLAN

This document serves as the written Chemical Hygiene Plan (CHP) for Pathology laboratories using chemicals at Royal Hospital. The CHP is a regular, continuing effort, not a standby or short term activity. Departments, divisions, sections, or other work units engaged in laboratory work whose hazards are not sufficiently covered in this written manual must customize it by adding their own sections as appropriate (e.g. standard operating procedures, emergency procedures, identifying activities requiring prior **approval before commencing work.**

RELATED DOCUMENTS

HSE Induction – New Staff

HSE Audit template

Technical Procedure Risk assessment

COSHH risk assessment template

Waste Management Audit checklist

Chemical Hygiene Plan

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